

Name _____ Surname _____ Date ____/____/____

Midterm test, Term 1 — Speed, energy and work

Version B

Time: 75 minutes. Calculator and formula sheet (last page) allowed. Use $g = 10 \text{ N/kg}$. Write on separate sheets, show every step of your working and write the unit in each answer. Give answers to 3 significant figures where needed. Total: 55 marks.

 **Lifebuoy.** The parts marked with the lifebuoy are worth 33 marks out of 55, that is 60%: if you have completed all of them, you are safe. The other parts take you beyond the pass mark.

Exercise 1 — Multiple choice

 Only one answer is correct. Each question is worth 1 mark, 8 marks in total.

1. A ball is held above the ground. Its energy is mainly stored as:
A kinetic energy B elastic (strain) energy C gravitational potential energy D chemical energy
2. A cyclist rides 45 km in 1.5 h. Her average speed is:
A 30 km/h B 0.033 km/h C 67.5 km/h D 46.5 km/h
3. A flat, horizontal line on a distance–time graph shows an object that is:
A moving at constant speed B stationary C speeding up D moving backwards
4. A pendulum bob swings down from its highest point. Ignoring air resistance, which quantity stays constant?
A the kinetic energy B the gravitational potential energy C the speed D the total energy
5. The speed of a car triples. Its kinetic energy becomes:
A three times bigger B six times bigger C nine times bigger D the same
6. In which situation is work being done?
A a book rests on a table B a man lifts a suitcase onto a shelf C a child sits on a chair
D a student holds a heavy box still
7. A force of 20 N moves a box 0.50 m in the direction of the force. The work done is:
A 10 J B 40 J C 20.5 J D 0.025 J
8. At the highest point of a swing a girl has 300 J of gravitational potential energy. At the lowest point, ignoring friction, her kinetic energy is:
A 150 J B 0 J C 600 J D 300 J

Exercise 2 — True or false

 Write T or F. No explanation needed. Each statement is worth 1 mark, 4 marks in total.

- (a) A flat horizontal line on a distance–time graph shows an object moving at constant speed.
- (b) A box is pushed along a level floor: its gravitational potential energy stays the same.
- (c) The kinetic energy of an object depends on its mass and on its speed.
- (d) Work done is measured in newtons.

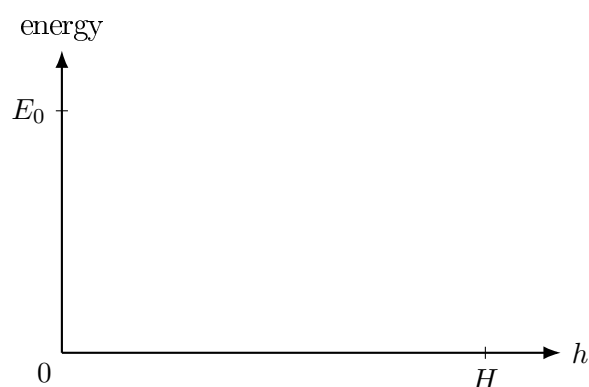
Exercise 3 — Energy stores

⚽ Complete the table. In each event, name the energy store that decreases and the one that increases. 1 mark for each row, 3 marks in total.

Event	Store that decreases	Store that increases
A trolley rolls down a ramp (ignore friction).		
A stretched elastic band shoots a paper ball.		
A crane lifts a load, using fuel (ignore thermal energy).		

Exercise 4 — A ball thrown upwards

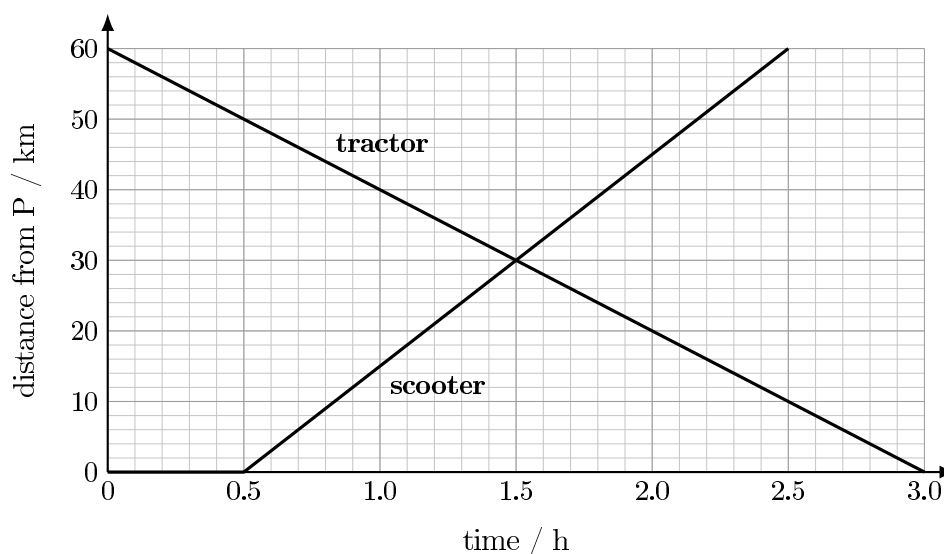
A ball of mass 0.40 kg is thrown vertically upwards from the ground at 15 m/s. Ignore air resistance. On the axes below, E_0 is the kinetic energy at launch and H is the maximum height.



- ⚽ Draw and label how the kinetic energy and the gravitational potential energy change with the height h , from $h = 0$ to $h = H$. [2]
- ⚽ Write, in symbols and without numbers, the equation that gives the maximum height H . [2]
- Calculate the kinetic energy at launch. [2]
- Calculate the maximum height. [2]
- ⚽ The same object is thrown with twice the speed. The maximum height becomes: the same, twice as big, four times as big, eight times as big. Circle the correct answer and give a reason in one line. [2]

Exercise 5 — Two travellers

A tractor and a scooter travel on the same straight road between town P and town Q. The graph shows the distance of each of them from P. The tractor starts from Q and the scooter starts from P at the same moment.



- (a) Read the graph: for how long does the scooter stay at rest at the start, and at what time and at what distance from P do the two meet? [2]
- (b) Calculate the speed of the scooter while it is moving, in km/h. [2]
- (c) Calculate the average speed of the scooter for the whole journey, including the rest, in km/h. [2]
- (d) Calculate the speed of the tractor in m/s. [2]

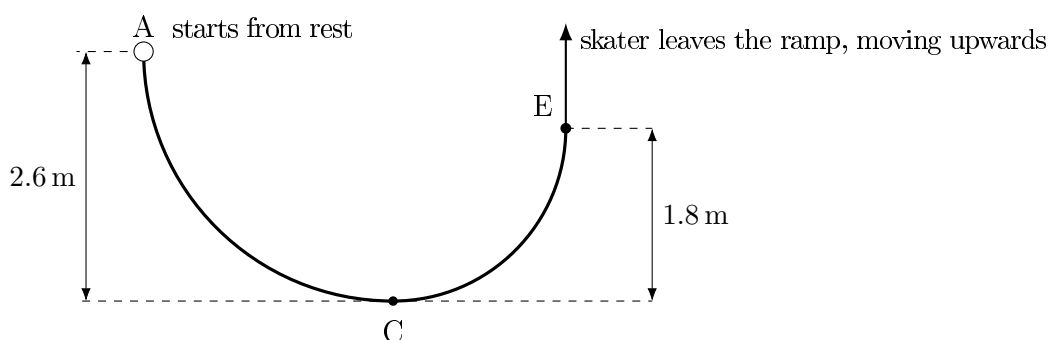
Exercise 6 — A van brakes

A van of mass 2000 kg travels at 12 m/s. The driver brakes and the van stops after 30 m. Assume that the braking force is constant.

- (a) In which energy store does the kinetic energy mainly end up? [1]
- (b) Calculate the kinetic energy before braking. [2]
- (c) Calculate the braking force. [2]
- (d) The van now travels at half the speed, with the same mass and the same braking force. The stopping distance becomes: the same, half as long, one quarter as long, one eighth as long. Circle the correct answer and give a reason in one line. [1]

Exercise 7 — The half-pipe

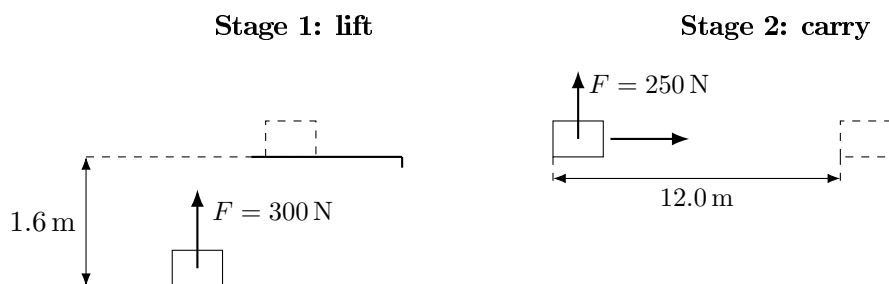
A skateboarder of mass 60 kg starts from rest at A, at the top of a half-pipe, 2.6 m above the lowest point C. He rolls down, passes C and goes up the right-hand side, leaving the ramp at E, 1.8 m above C, moving vertically upwards. Ignore friction and air resistance.



- Without calculating, say how high above C the skateboarder rises before he starts to fall, and justify your answer in one line. [2]
- Write, in symbols and without numbers, the equation that gives his speed v at E in terms of the heights h_A and h_E above C. [2]
- Calculate the gravitational potential energy lost between A and E. [2]
- Calculate his speed at E. [2]

Exercise 8 — Lifting and carrying a box

A warehouse worker lifts a box of mass 25 kg from the floor to a high shelf 1.6 m above it (stage 1). His hands push upwards on the box with a constant force of 300 N, and the box is still moving upwards when it reaches the height of a high shelf. In stage 2 he walks 12.0 m along an aisle, holding the box at that height with a force of 250 N from his hands, pointing upwards.



- Write, in symbols, the equation for the work done by his hands and the equation for the gain in gravitational potential energy of the box in stage 1. [2]
- Calculate the work done by his hands on the box and the gain in gravitational potential energy of the box in stage 1. [2]
- The work done is greater than the gain in gravitational potential energy. In which energy store does the difference end up? [1]
- State the work done by his hands on the box in stage 2. [1]
- Explain your answer to (d) in one or two lines. (His muscles use chemical energy even while he carries the box: here we ask only about the work done on the box.) [2]